

# Lokesh Miriyala

lokeshmiriyala11@gmail.com | (959) 221-9158 | USA | LinkedIn

## Summary

AI/ML Engineer with 3+ years of strong expertise in building scalable machine learning systems, developing NLP models, and deploying solutions across cloud platforms. Skilled in data preprocessing, model optimization, and MLOps practices. Adept at working with cross-functional teams to deliver high-impact, production-ready AI solutions that align with business and compliance requirements.

## Technical Skills

- Programming Languages:** Python (BeautifulSoup, Pydantic, pandas, spaCy), R, Julia, SQL, PySpark, Bash, Java, C++, Shell scripting
- Machine Learning & Deep Learning:** TensorFlow, PyTorch, Scikit-learn, Keras, Hugging Face (Transformers, Accelerate), OpenAI GPT, XGBoost, LightGBM, CatBoost, Parameter-Efficient Fine-Tuning (PEFT), DeepSpeed, transfer learning
- Deep Learning Architectures:** CNN, RNN, LSTM, Transformers (BERT, RoBERTa, DeBERTa), GANs, Graph Neural Networks
- Natural Language Processing (NLP):** spaCy, NLTK, LangChain, FAISS, Named Entity Recognition (NER), Sentiment Analysis, Text Classification, Word Embeddings (Word2Vec, GloVe), Topic Modeling, TF-IDF, Clinical NLP
- Data Engineering & Pipeline Tools:** Apache Spark, Hadoop, Dask, PySpark, Apache Airflow, Azure Data Factory, Azure Databricks, AWS Glue, SQLAlchemy, Kafka, Redis, ETL pipeline design
- Big Data & Streaming Technologies:** Apache Kafka, Apache Flink, AWS Kinesis, Google Pub/Sub
- Cloud Platforms & Infrastructure:** AWS (SageMaker, Lambda, EC2, Elastic Beanstalk, EKS), Microsoft Azure (Machine Learning, Kubernetes), GCP (AI Platform, BigQuery), Docker, Kubernetes, Terraform, gRPC, containerization, Rasa, Pinecone, real-time chatbot workflows
- Model Optimization & Hyperparameter Tuning:** Optuna, Grid Search, Bayesian Optimization, Hyperopt, k-Fold Cross-Validation
- Model Monitoring & MLOps:** MLflow, GitLab CI/CD, Jenkins, Evidently, Prometheus, Grafana, Drift Detection, DVC
- Visualization & Reporting:** Streamlit, Plotly, Matplotlib, Seaborn, Tableau, Power BI, Looker
- Version Control & Collaboration:** Git, GitHub, GitLab, Bitbucket
- Databases & APIs:** SQL/NoSQL databases (PostgreSQL, MongoDB), REST APIs, Microservices Architecture
- Security & Compliance:** Data Privacy & Security (SOC 2, PCI DSS compliance)
- Explainable AI:** SHAP, LIME
- Methodologies & Soft Skills:** Agile & Scrum methods, Analytical Thinking, Problem Solving, Collaboration, Communication, Ethical AI Practices

## Professional Experience

### AI/ML Engineer, TD Bank

10/2024 – Present | Remote, USA

- Developed an AI-driven personalized client engagement assistant that enhanced user interaction by 22% and increased conversion metrics by 17%, team up with product, data science, risk, and compliance teams to define structured requirements aligned with TD's customer-centric approach.
- Designed and implemented scalable, privacy-compliant ETL pipelines leveraging Apache Airflow, Python, and AWS Glue to process millions of client interactions, transaction events, and portfolio updates, ensuring timely, secure, and auditable data flows for downstream AI/ML models within TD's data governance frameworks.
- Engineered advanced customer behavior features using spaCy, pandas, and scikit-learn to analyze client intent, generate session embeddings, and detect financial lifecycle patterns, improving feature robustness and reducing cold-start issues by 30%.
- Fine-tuned transformer-based NLP models (BERT, DeBERTa, and domain-specific variants) on AWS SageMaker using labeled financial transaction records, client feedback, and search logs, significantly boosting personalized recommendation accuracy (NDCG@10 from 0.62 to 0.84).
- Applied Parameter-Efficient Fine-Tuning (PEFT) techniques such as LoRA with Hugging Face Accelerate and DeepSpeed on AWS EC2 to adapt large language models efficiently within TD Bank's strict data governance, security, and privacy policies.
- Utilized Optuna for sophisticated hyperparameter optimization and managed over 500 training experiments via MLflow, creating a reproducible and auditable model registry to support rigorous model validation and regulatory compliance.
- Collaborated with DevOps teams to containerize AI microservices with Docker, deploying on AWS EKS clusters, and exposing APIs through TD Bank's secure GraphQL gateway. Authored detailed API documentation, versioning protocols, and performance metrics for continuous system monitoring and compliance auditing.

### AI/ML Engineer, The Cigna Group

01/2021 – 07/2023 | Chennai, India

- Directed the development of Claims Analytics Platform automating classification, anomaly detection, and fraud flagging. Improved processing speed by 18% and reduced fraudulent claims by 12%. Collaborated with Compliance, Claims, and Fraud teams during requirements gathering.
- Developed robust ETL pipelines on Azure Data Factory using Python, SQL, and PySpark to preprocess structured claims data, billing codes, and unstructured physician notes, ensuring high-quality, labeled datasets for model training and analysis.
- Built classification and anomaly detection models using scikit-learn, XGBoost, and autoencoders to accurately categorize claims types and flag high-risk cases achieving 87% F1-score in classification and 92% recall in fraud detection.
- Performed hyperparameter tuning on XGBoost and BERT models using Grid Search on Azure ML to optimize performance and used SHAP explainability for transparent model interpretation to increase stakeholder trust in predictions.
- Extracted clinical insights from unstructured physician notes using BERT and TF-IDF with logistic regression deployed on Azure ML, improving claim urgency detection and clinical tagging to support faster decision-making and reduce manual review.
- Containerized models with Docker and deployed on Kubernetes clusters on Azure, exposing REST APIs for real-time inference integrated directly into Cigna's claims processing platform, reducing manual review workload by 25%.
- Scheduled monthly retraining workflows using Apache Airflow on Azure, monitored model drift and performance via Prometheus and Grafana dashboards, ensuring consistent accuracy, fairness, and compliance with regulatory requirements.

## Education

Master of Science in Computer Science

08/2023 – 04/2025

Rivier University - Nashua, NH, USA

Bachelor of Engineering in Electronics and Communication Engineering

08/2019 – 05/2023

Bharath Institute of Higher Education and Research - Chennai, India